

I&S-ViT: An Inclusive & Stable Method for Post-Training ViTs Quantization

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Abstract—Albeit the scalable performance of vision transformers (ViTs), the dense computational costs undermine their position in industrial applications. Post-training quantization (PTQ), tuning ViTs with a tiny dataset and running in a low-bit format, well addresses the cost issue but unluckily bears more performance drops in lower-bit cases. In this paper, we introduce I&S-ViT, a novel method that regulates the PTQ of ViTs in an inclusive and stable fashion. I&S-ViT first identifies two issues in the PTQ of ViTs: (1) Quantization inefficiency in the prevalent log2 quantizer for post-Softmax activations; (2) Rugged and magnified loss landscape in coarse-grained quantization granularity for post-LayerNorm activations. Then, I&S-ViT addresses these issues by introducing: (1) A novel shift-uniform-log2 quantizer (SULQ) that incorporates a shift mechanism followed by uniform quantization to achieve both an inclusive domain representation and accurate distribution approximation; (2) A three-stage smooth optimization strategy (SOS) that amalgamates the strengths of channel-wise and layer-wise quantization to enable stable learning. Comprehensive evaluations across diverse vision tasks validate I&S-ViT's superiority over existing PTQ of ViTs methods, particularly in low-bit scenarios. For instance, I&S-ViT elevates the performance of W3A3 ViT-B by an impressive 50.68%.

Index Terms—Network quantization, post-training quantization, vision transformers.

Received 15 February 2024; revised 20 August 2025; accepted 9 September 2025. Date of publication 16 September 2025; date of current version 9 January 2026. This work was supported in part by the National Key R&D Program of China under Grant 2022ZD0118202, in part by the National Science Fund for Distinguished Young Scholars under Grant 62025603, in part by the National Natural Science Foundation of China under Grant U21B2037, Grant U22B2051, Grant 62176222, Grant 62176223, Grant 62176226, Grant 62072386, Grant 62072387, Grant 62072389, Grant 62002305, and Grant 62272401, and in part by the Natural Science Foundation of Fujian Province of China under Grant 2021J01002 and Grant 2022J06001. Recommended for acceptance by R. K. Iyer. (Corresponding author: Rongrong Ji.)

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Our code is available at <https://github.com/zysxmu/IaS-ViT>.

Digital Object Identifier 10.1109/TPAMI.2025.3610466

I. INTRODUCTION

IN the ever-evolving realm of computer vision, vision transformers (ViTs) of late [1] stand out as an excellent architecture to capture the long-range relationships among image patches with the multi-head self-attention (MHSA) mechanism. However, exceptional power comes at the expense of great computing: n image patches result in $\mathcal{O}(n^2)$ complexity from the MHSA operation. In order to provide affordable usage of ViTs, researchers from the vision community have strained every nerve to reduce the compute costs [2], [3], [4], [5], [6].

Model quantization reduces the representation precision of weights & activations, and has garnered sustainable attention due mostly to its reliable academic support and applied industrial practice [7]. A multitude of studies [8], [9], [10], [11] have run into quantization-aware training (QAT) by accessing the entire training dataset and executing an end-to-end retraining. Such premises require a very dense computational cost in network retraining, which sadly drops an obstacle to the broad deployment of QAT methods. Therefore, researchers have gravitated to post-training quantization (PTQ) in search of quantizing models with a tiny dataset, for the sake of minor costs [4], [12], [13], [14], [15]. To adapt to the specific structure in ViTs such as LayerNorm and self-attention mechanisms, current efforts on PTQ of ViTs typically introduce dedicated quantizers and quantization schemes to maintain ViTs' original performance. For example, FQ-ViT [4] and PTQ4ViT [16] respectively introduce a log2 quantizer and a twin uniform quantizer for post-Softmax activations. RepQ-ViT [2] adopts the channel-wise quantizer for high variant post-LayerNorm activations first and then reparameterizes it to a layer-wise quantizer. Notwithstanding, considerable performance drops are observed when performing low-bit quantization. By way of illustration, in the W4A4 case, RepQ-ViT [2] causes 10.82% accuracy drops over full-precision DeiT-S [17] on ImageNet [18]; while in W3A3, it leads to 74.48% accuracy drops. Recent optimization-based PTQ methods have demonstrated their capacity in quantizing convolutional neural networks (CNNs) [19], [20], [21]. However, their attempts in ViTs remain unexploited, and in Table I of this paper, we find their applications typically result in overfitting in high-bit cases and suffer large performance degradation in ultra-low bit cases, which in turn, barricades their capacity in ViTs architectures [2], [22], [23], [24].

In this paper, we present a novel optimization-based PTQ method specifically tailored for ViTs, called I&S-ViT, to

TABLE I

QUANTIZATION RESULTS ON IMAGENET DATASET. THE TOP-1 ACCURACY (%) IS REPORTED AS THE METRIC. "OPTI." DENOTES THE OPTIMIZATION-BASED METHOD, "BIT. (W/A)" INDICATES THAT THE BIT-WIDTH OF THE WEIGHTS AND ACTIVATIONS IS W AND A BITS, RESPECTIVELY. "†" INDICATES THE RESULTS ARE REPRODUCED FROM THE OFFICIAL CODES

Method	Opti.	Bit. (W/A)	ViT-S	ViT-B	DeiT-T	DeiT-S	DeiT-B	Swin-S	Swin-B
Full-Precision	-	32/32	81.39	84.54	72.21	79.85	81.80	83.23	85.27
FQ-ViT [†] [4]	$\times$	3/3	0.10	0.10	0.10	0.10	0.10	0.10	0.10
PTQ4ViT [†] [16]	$\times$	3/3	0.01	0.01	0.04	0.01	0.27	0.35	0.29
BRECQ [†] [19]	$\checkmark$	3/3	0.42	0.59	25.52	14.63	46.29	11.67	1.7
QDrop [†] [20]	$\checkmark$	3/3	4.44	8.00	30.73	22.67	24.37	60.89	54.76
PD-Quant [†] [21]	$\checkmark$	3/3	1.77	13.09	39.97	29.33	0.94	69.67	64.32
GPTQ [†] [53]	$\times$	3/3	1.17	0.28	1.72	5.89	27.95	27.24	11.14
RepQ-ViT [†] [2]	$\times$	3/3	0.43	0.14	0.97	4.37	4.84	8.84	1.34
I&S-ViT (Ours)	$\checkmark$	3/3	45.16	63.77	41.52	55.78	73.30	74.20	69.30
FQ-ViT [4]	$\times$	4/4	0.10	0.10	0.10	0.10	0.10	0.10	0.10
PTQ4ViT [16]	$\times$	4/4	42.57	30.69	36.96	34.08	64.39	76.09	74.02
APQ-ViT [22]	$\times$	4/4	47.95	41.41	47.94	43.55	67.48	77.15	76.48
BRECQ [†] [19]	$\checkmark$	4/4	12.36	9.68	55.63	63.73	72.31	72.74	58.24
QDrop [†] [20]	$\checkmark$	4/4	21.24	47.30	61.93	68.27	72.60	79.58	80.93
PD-Quant [†] [21]	$\checkmark$	4/4	1.51	32.45	62.46	71.21	73.76	79.87	81.12
GPTQ [†] [53]	$\times$	4/4	67.60	75.00	58.69	70.88	76.21	80.11	81.06
RepQ-ViT [2]	$\times$	4/4	65.05	68.48	57.43	69.03	75.61	79.45	78.32
I&S-ViT (Ours)	$\checkmark$	4/4	74.87	80.07	65.21	75.81	79.97	81.17	82.60
FQ-ViT [†] [4]	$\times$	5/5	0.10	0.10	0.10	0.10	0.10	0.10	0.10
PTQ4ViT [†] [16]	$\times$	5/5	72.63	72.23	65.38	71.13	72.32	81.17	81.59
BRECQ [†] [19]	$\checkmark$	5/5	50.59	53.72	63.97	67.49	75.87	81.06	82.48
QDrop [†] [20]	$\checkmark$	5/5	56.46	57.90	64.61	71.27	78.89	81.53	84.01
PD-Quant [†] [21]	$\checkmark$	5/5	65.78	60.31	68.94	75.11	74.87	81.76	82.65
GPTQ [†] [53]	$\times$	5/5	78.61	82.07	69.10	77.11	80.18	82.05	83.03
RepQ-ViT [†] [2]	$\times$	5/5	78.43	82.03	69.00	77.04	80.08	82.00	83.22
I&S-ViT (Ours)	$\checkmark$	5/5	79.34	83.03	69.78	78.33	81.22	82.07	84.05
FQ-ViT [4]	$\times$	6/6	4.26	0.10	58.66	45.51	64.63	66.50	52.09
PSAQ-ViT [13]	$\times$	6/6	37.19	41.52	57.58	63.61	67.95	72.86	76.44
Ranking-ViT [24]	$\checkmark$	6/6	-	75.26	-	74.58	77.02	-	-
EasyQuant [67]	$\checkmark$	6/6	75.13	81.42	-	75.27	79.47	82.45	84.30
PTQ4ViT [16]	$\times$	6/6	78.63	81.65	69.68	76.28	80.25	82.38	84.01
APQ-ViT [22]	$\times$	6/6	79.10	82.21	70.49	77.76	80.42	82.67	84.18
NoisyQuant-Linear [14]	$\times$	6/6	76.86	81.90	-	76.37	79.77	82.78	84.57
NoisyQuant-PTQ4ViT [14]	$\times$	6/6	78.65	82.32	-	77.43	80.70	82.86	84.68
BRECQ [†] [19]	$\checkmark$	6/6	54.51	68.33	70.28	78.46	80.85	82.02	83.94
QDrop [†] [20]	$\checkmark$	6/6	70.25	75.76	70.64	77.95	80.87	82.60	84.33
PD-Quant [†] [21]	$\checkmark$	6/6	70.84	75.82	70.49	78.40	80.52	82.51	84.32
Bit-shrinking [23]	$\checkmark$	6/6	80.44	83.16	-	78.51	80.47	82.44	-
GPTQ [†] [53]	$\times$	6/6	80.43	83.71	70.80	78.92	81.38	82.80	84.69
RepQ-ViT [2]	$\times$	6/6	80.43	83.62	70.76	78.90	81.27	82.79	84.57
I&S-ViT (Ours)	$\checkmark$	6/6	80.43	83.82	70.85	79.15	81.68	82.89	84.94

harness the potential of optimization-based techniques. At first, we identify that the log2 quantizer, widely adopted for long-tailed post-Softmax activations, suffers from the quantization inefficiency issue, which refers to the representative range failing to encompass the entire input domain. In response, we propose a shift-uniform-log2 quantizer (SULQ). This novel quantizer, by introducing an initial shift bias to the log2 function input, subsequently uniformly quantizes its outputs. SULQ is able to fully include the input domain to solve the quantization inefficiency issue and accurately approximate the distribution of post-Softmax activations. Moreover, SULQ can be efficiently executed by the fast and hardware-friendly bit-shifting operations [2], [4].

Furthermore, we observe marked distinctions in the loss of landscapes across different quantization granularities. As shown in Fig. 4, channel-wise weight quantization and layer-wise

post-LayerNorm activation quantization result in a rugged and magnified loss, impeding quantization learning and compromising model performance [15], [25], [26]. This aggravation can be alleviated by executing full-precision weights. Further applying channel-wise quantization to post-LayerNorm activations results in a smooth landscape with reduced loss magnitudes, leading to more stable and effective optimization [23], [27]. Motivated by these insights, we propose a three-stage smooth optimization strategy (SOS) to harness the benefits of the smooth and low-magnitude loss landscape for optimization, while maintaining the efficiency of the layer-wise quantization for activations [2], [7], [28]. In the first stage, we fine-tune the model with full-precision weights alongside channel-wise quantized post-LayerNorm activations, and other activations employ a layer-wise quantizer. In the second stage, we seamlessly transit the channel-wise quantizer to its layer-wise counterpart with

the scale reparameterization technique [2]. Finally, in the third stage, the model undergoes fine-tuning with both activations and weights subjected to quantization for restoring the performance degradation of weights quantization.

Comprehensive experimental assessments across a wide range of ViT variants and vision tasks validate the preeminence of the proposed I&S-ViT. For instance, for the W3A3 ViT-B, I&S-ViT significantly elevates performance, registering an encouraging improvement of 50.68%.

II. RELATED WORK

A. Vision Transformers (ViTs)

Subsequent to CNNs, ViTs [1] have again revolutionized the field of computer vision. ViTs tokenize an image as the input of a transformer architecture [29], therefore, a structured image is processed in a sequence fashion. Given that the performance of vanilla ViTs relies on the large-scale pre-trained dataset, DeiT [17] develops an efficient teacher-student training approach. In addition to image classification, ViTs have been well adopted in low-level vision [30] and video process [31], etc. Liang et al. [30] proposed SwinIR that builds on the Swin transformer block to solve image restoration tasks. In [31], a pure-transformer model is proposed for video classification, wherein spatio-temporal tokens from videos are encoded using a series of transformer layers. In particular, Swin's hierarchical structure with the shifted window-based self-attention [32], extends ViTs' applicability to dense vision tasks such as object detection [33], [34] and segmentation [35]. However, the impressive performance of ViTs relies on significant computational overhead, making them challenging for resource-constrained environments [36], [37].

B. ViTs Quantization

By reducing the numerical precision, model quantization has been instrumental in providing deployment for neural networks [7]. Model quantization techniques are commonly categorized into three types: quantization-aware training (QAT), post-training quantization (PTQ), and data-free quantization (DFQ). QAT methods [8], [9], [10], [38], [39], [40], [41], [42] tend to achieve the best performance, as they allow the model to be retrained to adapt to quantization. However, they come with significant drawbacks, including the need for full training data and the computational burden of retraining. In contrast, DFQ methods [13], [43], [44], [45], [46], [47], [48], [49], [50] eliminate the need for training data by synthesizing pseudo samples based on prior guidance. While data-independent, these methods often suffer from noticeable performance degradation due to the lack of real data for model adaptation. Given these trade-offs, recent research has increasingly focused on PTQ for ViTs, due to its favorable balance between accuracy, minimal data requirements, and ease of deployment in practical settings [12], [51]. Unfortunately, the customized operators, such as LayerNorm and MHSA in ViTs, create maladjustments when making a direct extension of PTQ methods from CNNs to ViTs [2], [4], [19], [20], [52].

Consequently, there is a growing consensus to develop ViTs-specialized PTQ methods. Liu et al. [24] introduce the first PTQ method specifically designed for ViTs. They develop a ranking loss to maintain the relative order of the post-Softmax activations. Moreover, they adopt a nuclear norm-based mixed-precision scheme to enable a mix-precision assignment for exploiting various quantization sensitivities of different layers. FQ-ViT [4] introduces a fully-quantized method for ViTs, incorporating Powers-of-Two Scale and Log-Int-Softmax for Layer-Norm and post-Softmax activations, respectively. PTQ4ViT [16] adopts a twin uniform quantization method to reduce the quantization error on the long-tail post-Softmax activations and uneven post-GELU activation. In addition, a Hessian-guided metric is used for searching quantization scales. Liu et al. [14] theoretically prove that adding uniform noise can reduce the quantization error, and thus suggest applying the uniform noise bias to activations to improve the performance of quantized ViTs. APQ-ViT [22] establishes a calibration strategy that considers the block-wise quantization error and proposes a Matthew-effect preserving quantizer for post-Softmax activations. Evol-Q [15] argues that the quantization parameters incur an extremely ragged loss landscape and thus adopts an evolutionary search to determine the disturbance-sensitive quantization parameters. Bit-shrinking [23] proposed gradually decreasing the bit-width to achieve a good initialization point. RepQ-ViT [2] first deploys complex quantizers for post-LayerNorm activations, subsequently simplifying these quantizers through parameterization. Moreover, the $\text{Log}\sqrt{2}$ quantizer that allocates more bits on the near-one region is adopted for post-Softmax activations. GPTQ [53] employs OBS [54] to progressively quantize part of the full-precision weight and update the remaining weights to compensate for the quantization error. Recently, ERQ [55] proposes an error reduction-based method, which utilizes the classical ridge regression to achieve a fast and effective quantization. APHQ-ViT [56] and FIMA-Q [57] respectively employ Average Perturbation Hessian and Fisher Information Matrix to guide the reconstruction process. OASPTQ [58] presents a theoretical analysis of reconstruction granularity in mitigating the outlier problem in ViTs. Despite current PTQ methods of ViTs having made significant progress, these methods still suffer from non-trivial performance degradation, especially in ultra-low bit cases. Addressing the performance drops remains a daunting task in the field.

III. PRELIMINARIES

Structure of ViTs: An input image I is first split into N flattened 2D patches, which are then projected by an embedding layer to D -dimensional vectors, denoted as $\mathbf{X}_0 \in \mathbb{R}^{N \times D}$. Then, $\mathbf{X}_0$ is fed into L transformer blocks, each of which consists of a multi-head self-attention (MHSA) module and a multi-layer perceptron (MLP) module. For the l -th transformer blocks, the computation can be expressed as:

$$\mathbf{Z}_{l-1} = \text{MHSA}_l(\text{LayerNorm}(\mathbf{X}_{l-1})) + \mathbf{X}_{l-1}. \quad (1)$$

$$\mathbf{X}_l = \text{MLP}_l(\text{LayerNorm}(\mathbf{Z}_{l-1})) + \mathbf{Z}_{l-1}. \quad (2)$$

MHSA consists of H self-attention heads. For the h -th head, the operations with input $\mathbf{X}_{l-1,h}$ formulated below:

$$[\mathbf{Q}_h, \mathbf{K}_h, \mathbf{V}_h] = \mathbf{X}_{l-1,h} \mathbf{W}_h^{QKV} + \mathbf{b}_h^{QKV}. \quad (3)$$

$$\mathbf{A}_h = \text{Softmax} \left(\frac{\mathbf{Q}_h \cdot \mathbf{K}_h^T}{\sqrt{D_h}} \right) \mathbf{V}_h, \quad (4)$$

where D_h is the dimension size of each head. Denoting $\mathbf{X}_{l-1} = \text{concat}(\mathbf{X}_{l-1,1}, \mathbf{X}_{l-1,2}, \dots, \mathbf{X}_{l-1,H})$, the results of each head are concatenated and the output of the l -th MHSA is obtained by:

$$\text{MHSA}(\mathbf{X}_{l-1}) = \text{concat}(\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_H) \mathbf{W}^O + \mathbf{b}^O. \quad (5)$$

The MLP module contains two fully-connected layers (FC) and the GELU activation function. Denoting the input to the l -th MLP module as $\mathbf{Z}_{l-1}$, the calculation is as:

$$\text{MLP}(\mathbf{Z}_{l-1}) = \text{GELU}(\mathbf{Z}_{l-1} \mathbf{W}^1 + \mathbf{b}^1) \mathbf{W}^2 + \mathbf{b}^2. \quad (6)$$

It can be seen that the major computation costs of ViTs come from the large matrix multiplications. Therefore, as a common practice in previous works [2], [16], we choose to quantize all the weights and inputs of matrix multiplications, leaving LayerNorm and Softmax operations as full-precision types.

Quantizers: The uniform quantizer evenly maps full-precision values $\mathbf{X}$ to integer $\mathbf{X}_q$. Given bit-width b , the uniform quantizer (UQ) is formally defined as:

$$\mathbf{X}_q = \text{UQ}(\mathbf{X}, b) = \text{clamp} \left(\left\lfloor \frac{\mathbf{X}}{s} \right\rfloor + z, 0, 2^b - 1 \right), \quad (7)$$

where $\lfloor \cdot \rfloor$ denotes the round function, clamp constrains the output between 0 and $2^b - 1$, s and z respectively are the quantization scale and the zero-point:

$$s = \frac{\max(\mathbf{X}) - \min(\mathbf{X})}{2^b - 1}, \quad z = \left\lfloor -\frac{\min(\mathbf{X})}{s} \right\rfloor. \quad (8)$$

Then, the de-quantized values $\bar{\mathbf{X}}$ can be calculated with de-quantization process D-UQ:

$$\bar{\mathbf{X}} = \text{D-UQ}(\mathbf{X}_q) = s(\mathbf{X}_q - z) \approx \mathbf{X}. \quad (9)$$

To handle the nature of the long-tail distribution of post-Softmax activations, the log2-based quantizer [59] has been extensively adopted in many previous PTQ methods of ViTs [2], [4], [15]. A common choice is using the log2 quantizer (LQ) for non-negative post-Softmax activation $\mathbf{X}$:

$$\mathbf{X}_q = \text{LQ}(\mathbf{X}, b) = \text{clamp} \left(\left\lfloor -\log_2 \frac{\mathbf{X}}{s} \right\rfloor, 0, 2^b - 1 \right). \quad (10)$$

Then, the de-quantization process D-LQ is used to obtain de-quantized values $\bar{\mathbf{X}}$:

$$\bar{\mathbf{X}} = \text{D-LQ}(\mathbf{X}_q) = s \cdot 2^{-\mathbf{X}_q} \approx \mathbf{X}. \quad (11)$$

For consistency with earlier works [2], [16], [22], we utilize the channel-wise quantizer for weights and the layer-wise quantizer for activations.

IV. METHOD

A. Block-Wise Optimization

In alignment with [19], [20], [22], we establish the block-wise reconstruction as the learning objective. Let $\mathbf{X}_l$ represent outputs of the l -th full-precision transformer block, and $\bar{\mathbf{X}}_l$ represent outputs of the quantized version. The block-wise reconstruction is defined as:

$$\mathcal{L}_l = \|\mathbf{X}_l - \bar{\mathbf{X}}_l\|_2. \quad (12)$$

Note that $\mathcal{L}_l$ is only backward to update weights in the l -th transformer block. In the next, we delve into the challenges and corresponding solutions. In Section IV-B, we first identify the quantization inefficiency issue of the log2 quantizer, and thus introduce our solution, i.e., shift-uniform-log2 quantizer. In Section IV-C, we find that scale smoothness varies across different quantization granularities, and thus propose our solution, i.e., smooth optimization strategy.

B. Shift-Uniform-Log2 Quantizer

The quantization of post-Softmax activations has an important effect on the performance of quantized ViTs [24]. As shown in Fig. 3, post-Softmax activations are non-negative, bounded within the range $[0, 1]$, and exhibit a highly skewed, long-tailed distribution with most values concentrated near zero. In Fig. 2(a), we plot the relationship of full-precision $\mathbf{X}$ and de-quantized $\bar{\mathbf{X}}$ when the uniform quantizer and log2 quantizer are deployed. The red line presents the quantized results of the uniform quantizer. It can be seen that the input is quantized uniformly. As a result, the near-zero values are all quantized to 0, which, however, fails to fit the distribution of long-tail post-Softmax activations. The blue line presents the quantized results of the log2 quantizer. Compared to the uniform quantizer, the log2 quantizer prioritizes more bits for the near-zero region, showing its advantage in addressing the prevalent long-tail distribution in post-Softmax activations [2], [4], [15], [60]. *However, log2 quantizer, as we analyze below, also exhibits a primary issue of quantization inefficiency.*

In Fig. 1(a), we give an example to elucidate what the issue is. Considering the input post-Softmax activations $\mathbf{X}$ with a range of $[1.08\text{e-}8, 0.765]$, the rounded results have a span of maximal 26 and minimal 0. However, the 3-bit quantization only covers a range of $[0, 7]$. Therefore, the rounded results lying in the segment $[8, 26]$ are subsequently clamped to 7. Note that the log quantizer designates the smallest integer to represent the value 0. As a result, in the 3-bit case, values yielding a rounded result within the range $[8, 26]$, are directly clamped to 0 after de-quantization. As for 4-bit quantization, the rounded segment $[16, 26]$ is clamped to 15. Consequently, values whose rounded result within the segment $[16, 26]$, are similarly clamped to 0. Consequently, many values of the near-zero region are clamped, resulting in range-uncovered de-quantized values. We name it “quantization inefficiency” in that a large portion of values are clamped to a position at remote. The post-Softmax activations have a plethora of zero-around values. The quantization inefficiency issue causes large quantization errors and compromises the model’s performance.

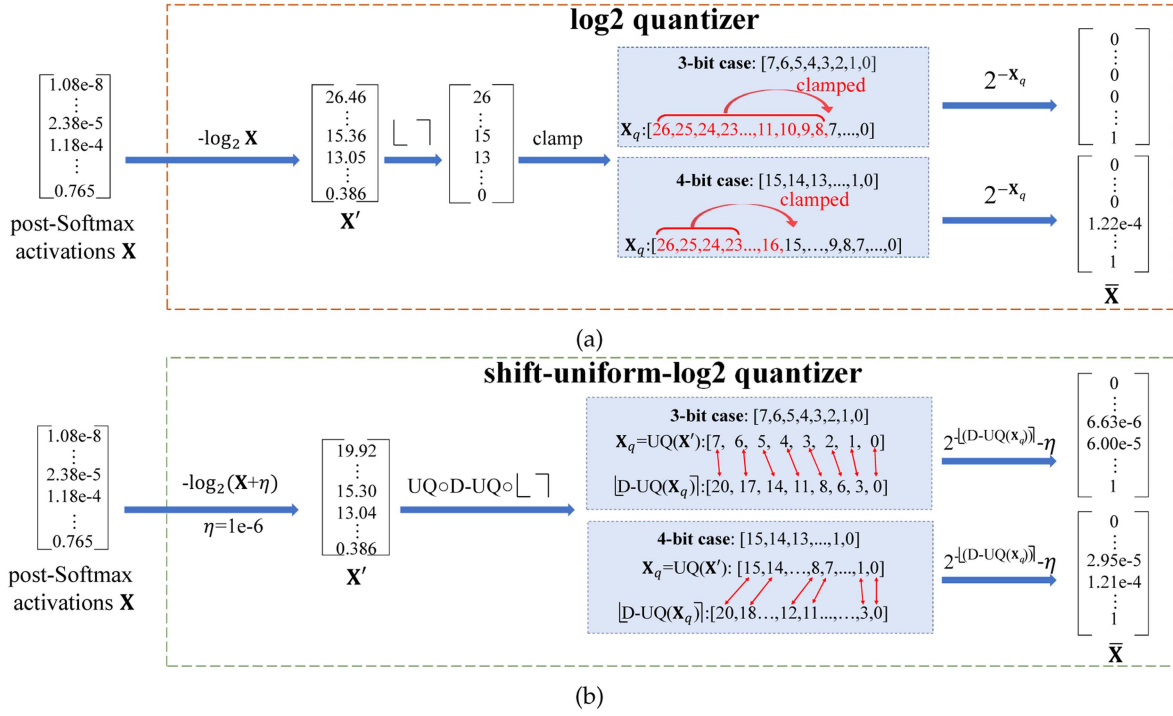


Fig. 1. (a) Illustration of the quantization inefficiency issue of the 3/4-bit log2 quantizers (LQ). For the 3-bit case, the values between 8 and 26 are clamped to 7. For the 4-bit case, the values between 16 and 26 are clamped to 15. Thus, the log2 quantizer results in range-uncovered de-quantized values. (b) Illustration of the quantization process of 3/4-bit shift-uniform-log2 quantizers (SULQ). SULQ exploits all bits to uniformly represent the range of input values, thereby resulting in range-covered de-quantized values.

Inspired by the above analyses, we introduce the shift-uniform-log2 quantizer (SULQ) to address the quantization inefficiency issue. Specifically, we first introduce a small positive bias term η to shift the full-precision input $\mathbf{X}$ before applying the base-2 logarithmic transformation. The shifted input is then passed through a uniform quantizer, resulting in the following quantization process:

$$\mathbf{X}_q = \text{SULQ}(\mathbf{X}, b) = \text{UQ}(-\log_2(\mathbf{X} + \eta), b), \quad (13)$$

where “UQ” denotes a uniform quantizer in Eq. (7). The de-quantization process of SULQ involves reversing the above steps: it first applies de-quantization of the uniform quantizer, then reverses the logarithmic transformation, and finally subtracts the shifting bias η . The process can be defined as:

$$\bar{\mathbf{X}} = \text{D-SULQ}(\mathbf{X}_q) = 2^{[-(D-UQ(\mathbf{X}_q))] - \eta} \approx \mathbf{X}, \quad (14)$$

“D-UQ” denotes the de-quantization process in (9) and $[\cdot]$ represents the round function. Note that the round function is applied to the outputs of D-UQ($\mathbf{X}_q$) to ensure integer outputs, such that fast and hardware-friendly bit-shifting operations can be applied [2], [4]. In our experiments, η is searched over the range $\{1e-6, 1e-5, 1e-4, 1e-3, 1e-2, 1e-1\}$. Fig. 2(b) presents the relationship of full-precision $\mathbf{X}$ and de-quantized $\bar{\mathbf{X}}$, w.r.t. different η , for ease of comparison with the uniform quantizer and log2 quantizer in Fig. 2(a). Also, Fig. 1(b) presents the 3/4-bit quantization processes of our SULQ. The proposed SULQ enjoys three advantages:

First, by exploiting the uniform quantizer, our SULQ effectively addresses the quantization inefficiency issue associated

with the log2 quantizer. In particular, the proposed SULQ inclusively represents the full range of the input domain. As illustrated in Fig. 1(b), for the 3-bit case, with an input range of $[0.386, 19.92]$, the uniform quantizer of SULQ evenly distributes the 8 integers to span the input range. That is, $[-(D-UQ(\mathbf{X}_q))]$ generates 8 values that evenly distribute over the input range of $[20, 0]$, encompassing 20, 17, 14, 11, 8, 6, 3, and 0, thus ensuring complete coverage of the input range of $[0.386, 19.92]$. Consequently, the de-quantized values of SULQ have a more inclusive range compared to those derived from the log2 quantizer. Similarly, for the 4-bit case, all 16 integers are employed to uniformly include the input range of $[0.386, 19.92]$. As shown in Fig. 1(b), SULQ yields an inclusive range and performs a more accurate quantization than the log2 quantizer. For example, for the 3-bit case, the log2 quantizer quantizes $1.18e-4$ and $2.38e-5$ to 0. In contrast, SULQ respectively quantizes them to $6.00e-5$ and $6.63e-6$. Clearly, SULQ yields a smaller quantization error.

Second, as shown in Fig. 2(b), SULQ implements a fine-grained quantization bit allocation strategy, which assigns more bits for regions close to zero and fewer bits for areas approaching one. This allocation paradigm is particularly effective in aligning with the long-tail distribution characteristic of post-Softmax activations.

Third, as revealed in Fig. 2(b), varying the parameter η leads to disparate quantization point distributions. Therefore, by adjusting η , SULQ can adapt to diverse input distributions. Moreover, given the same input range, the SULQ is able to produce more finer-grained representation as the bit increases. While the log2 quantizer allocates quantization points in a fixed pattern [38].

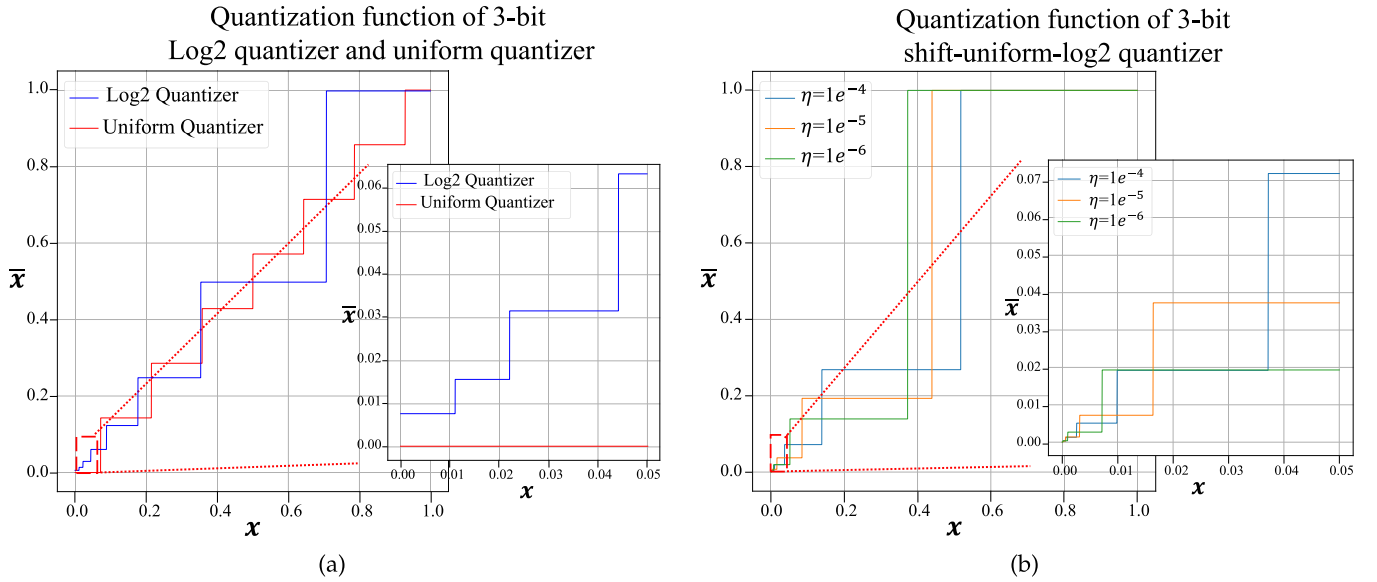


Fig. 2. Illustration of the quantization function of 3-bit (a) log2 quantizer and uniform quantizer. (b) shift-uniform-log2 quantizer.

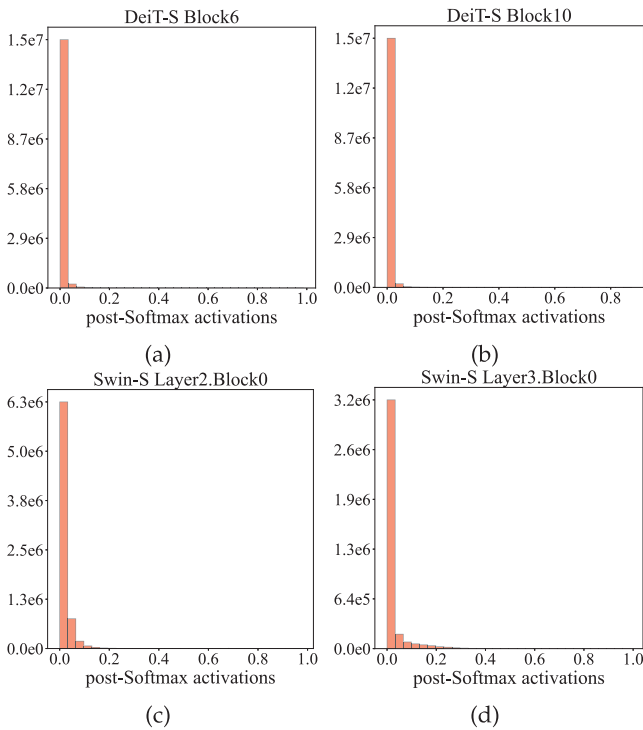


Fig. 3. Histogram of post-Softmax activations from Dei-T-S and Swin-S. The y-axis denotes the frequency. Note that the activations are non-negative.

Consequently, SULQ offers greater flexibility compared to the log2 quantizer.

Compared with the log2 quantizer, SULQ only involves one extra round function and two addition operations, the costs of which are negligible. Moreover, the uniform quantizer adopted in SULQ is well-supported by hardware and does not require any specialized components such as lookup tables (LUTs). During the inference, SULQ produces integer outputs. As a result, its

computations can be efficiently executed by fast and hardware-friendly bit-shifting operations, in line with previous works [2], [4], [15]. It is worth noting that many preceding methods perform transformations before executing uniform quantization, such as normalization [38], [61], [62] and power functions [63], [64]. However, these methods focus on weight quantization. In contrast, our SULQ is specifically tailored for post-Softmax activations by addressing the observed quantization inefficiency issue in the log2 quantizer, which remains largely untapped in prior research.

C. Smooth Optimization Strategy

It is a wide consensus that post-LayerNorm activations exhibit severe inter-channel variation, necessitating fine-grained quantization granularity [2], [60]. However, the effects of quantization granularity on the optimization process remain underexplored, and in this section, we intend to reveal the internal mechanism based on the loss landscape.

For demonstration, in Fig. 4, we present the loss landscape when post-LayerNorm activations are subjected to different quantization granularities. Following [15], we plot the loss landscape by adding perturbation to the model weights. Specifically, weights from two random channels are selected, and a basis vector is added to each.

As depicted in Fig. 4(a) and (b), if the weights undergo channel-wise quantization and post-LayerNorm activations undergo layer-wise quantization, the resulting landscape is rugged and magnified in its loss values. The rugged landscape can be attributed to the quantized weights. Specifically, distributing the quantized weights easily makes them transition to the next quantization point, causing a sudden and significant change in their values, resulting in jumps in the loss landscape [65]. Such an intricate and uneven landscape easily misdirects the learning path into a local minimum, which in turn compromises the

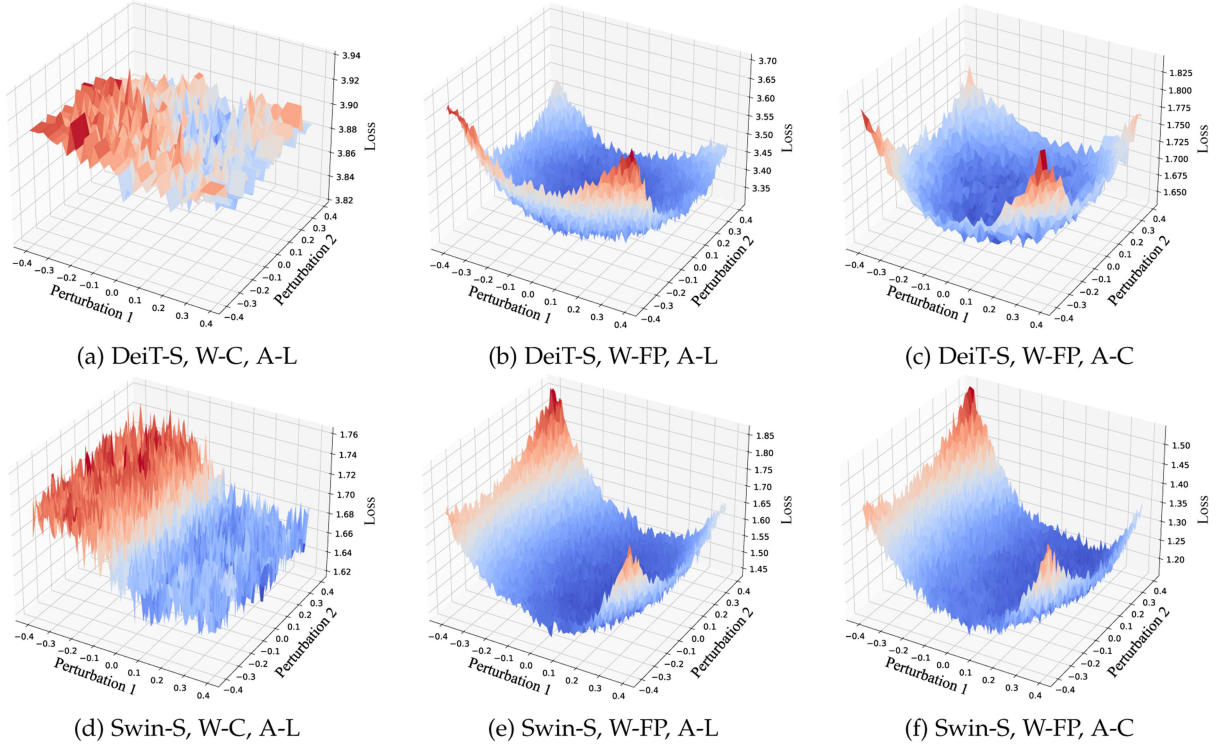


Fig. 4. Loss landscapes for W4A4 DeiT-S and Swin-S are visualized by perturbing weights along two basis vectors (Perturbation 1 & 2). The first row shows results of perturbing the qkv weights in Block10 of DeiT-S, and the second row shows results of perturbing the qkv weights in Layer1.Block1 of Swin-S. “W-C”, “W-FP” denote channel-wise weight quantization and full-precision weights, respectively. “A-L”, “A-C” denote layer-wise activation quantization and channel-wise activation quantization, respectively. (a) and (d): Channel-wise weight quantization & layer-wise activation quantization. (b) and (e): Full-precision weights & layer-wise activation quantization. (c) and (f): Full-precision weights & channel-wise activation quantization.

performance of quantized ViTs [15], [25], [26]. Motivated by this, we plot the loss landscape with full-precision weights. As shown in Fig. 4(b) and (e), maintaining weights at full-precision avoids the sudden change in weights, resulting in a significantly smoother loss landscape. However, there is still a high loss magnitude. The high magnitude of the loss landscape can be ascribed to the layer-wise quantization strategy, which utilizes identical quantization parameters across all channels. Such quantization parameters are ineffectual in accommodating the severe inter-channel variation in the post-LayerNorm activations, leading to substantial distortion in distribution [2], [4]. Fig. 4(c) and (f) plot the loss landscape when the post-LayerNorm activations are subjected to friendly channel-wise quantization. By employing this quantizer, the loss landscape is not just a gentle and even loss landscape, but one with reduced loss magnitude. Such a smooth and low-magnitude loss landscape reduces the learning difficulty [66], establishing a more secure and steadfast foundation upon which the optimization process can well proceed [23], [27].

Spurred by these insights, we introduce a training strategy, named smooth optimization strategy (SOS), to take advantage of the smooth and low-magnitude loss landscape for optimization at first, while afterward concurrently reaping the benefits of the efficiency proffered by the layer-wise quantizer [2], [7], [28]. The proposed SOS comprises three stages, as detailed below:

Stage one: To exploit the smooth and low-magnitude loss landscape as shown in Fig. 4(c) and (f), at this stage, we adopt

the channel-wise quantizer for post-LayerNorm activations, and the layer-wise quantizer for other activations. At the same time, we keep the network’s weights full-precision. We first calibrate the quantization parameters of activation quantizers by forwarding the calibration datasets once. Afterward, we optimize the model’s weights with the optimization objective in Section IV-A. With this setting, the optimization is performed with a smooth loss landscape with lower loss magnitude, thereby establishing a more secure and steadfast learning process.

Stage two: After stage one, the post-LayerNorm activations are channel-wise quantized, and other activations are layer-wise quantized. However, the channel-wise quantizer incurs extra overhead than the layer-wise. Therefore, we employ the scale reparameterization technique [2] to realize a transition from the channel-wise quantizer to its low-cost layer-wise equivalence. Specifically, given the channel-wise scales $s \in R^D$ and zero-point $z \in R^D$, $\tilde{s} = \text{Mean}(s) \in R^1$, $\tilde{z} = \text{Mean}(z) \in R^1$, $r_1 = s/\tilde{s}$, and $r_2 = z - \tilde{z}$. The reparameterization is completed by adjusting the LayerNorm’s affine parameters and the weights of the next layer of post-LayerNorm activations:

$$\tilde{\beta} = \frac{\beta + s \odot r_2}{r_1}, \quad \tilde{\gamma} = \frac{\gamma}{r_1}. \quad (15)$$

$$\tilde{W}_{:,j} = r_1 \odot W_{:,j}, \quad \tilde{b}_j = b_j - (s \odot r_2)W_{:,j}. \quad (16)$$

A detailed analysis can be found in [2]. It is worth noting that our strategy ensures a seamless transition while prior work

introduces a lossy transition. In [2], they remain the quantized weight $\mathbf{W}$. However, the calculation of $\mathbf{r}_1 \odot \mathbf{W}_{:,j}$ results in $\tilde{\mathbf{W}}$ with a changed distribution. As a result, the quantization parameters of $\mathbf{W}$ become unsuitable. In contrast, our strategy maintains weights at full-precision without quantization. Thus, the changed distribution resulting from $\mathbf{r}_1 \odot \mathbf{W}_{:,j}$ does not incur any mismatch.

Stage three: After stage two, activations of all layers are layer-wise quantized. Currently, the model's weights still remain full-precision. In this stage, we employ the channel-wise quantizer to quantize weights. At first, the quantization parameters of weight quantizers are calibrated by forwarding the calibration datasets once. Afterward, we again optimize the model's weight with the optimization objective in Section IV-A. This additional fine-tuning process with both quantized activations and weights further restores the performance degradation.

Analysis of SOS Stability: As illustrated in Fig. 4, applying channel-wise quantization to post-LayerNorm activations while keeping the weights in full precision results in a smoother and more stable loss landscape, compared to layer-wise activation quantization combined with quantized weights. We provide a formal justification by analyzing a simplified linear layer case, focusing on the Hessian matrix, a widely used indicator of loss flatness and optimization stability [19], [20], [66].

Let $\mathbf{W} \in \mathbb{R}^{m \times n}$ denote the full-precision weight matrix, and $\mathbf{X} \in \mathbb{R}^{n \times k}$ the full-precision activation. We define the channel-wise quantized version of $\mathbf{W}$ as $\mathbf{W}_{\text{cq}}$, and the channel-wise and layer-wise quantized versions of $\mathbf{X}$ as $\mathbf{X}_{\text{cq}}$ and $\mathbf{X}_{\text{lq}}$, respectively, with corresponding quantization errors δ_{cq} and δ_{lq} :

$$\mathbf{X}_{\text{cq}} = \mathbf{X} + \delta_{\text{cq}}, \quad \mathbf{X}_{\text{lq}} = \mathbf{X} + \delta_{\text{lq}}.$$

We consider the following two loss functions under different quantization schemes: $\mathcal{L}_1 = \|\mathbf{W}\mathbf{X} - \mathbf{W}_{\text{cq}}\mathbf{X}_{\text{lq}}\|_F^2$ for layer-wise quantized activation and channel-wise quantized weight, and $\mathcal{L}_2 = \|\mathbf{W}\mathbf{X} - \mathbf{W}\mathbf{X}_{\text{cq}}\|_F^2 = \|\mathbf{W}\delta_{\text{cq}}\|_F^2$ for channel-wise quantized activation with full-precision weights. To analyze the Hessian, we express the losses in vectorized form using the operator $\text{vec}(\cdot)$ and the Kronecker product $\otimes$. Let $w = \text{vec}(\mathbf{W})$, $w_c = \text{vec}(\mathbf{W}_{\text{cq}})$, and $C = \text{vec}(\mathbf{W}\mathbf{X})$. Define $P_1 = \mathbf{X}_{\text{lq}}^\top \otimes \mathbf{I}$ and $P_2 = \delta_{\text{cq}}^\top \otimes \mathbf{I}$. Then the losses can be rewritten as:

$$\mathcal{L}_1 = \|C - P_1 w_c\|_2^2, \quad (17)$$

$$\mathcal{L}_2 = \|P_2 w\|_2^2. \quad (18)$$

Since the Hessian of a quadratic form $\|y - Px\|_2^2$ is $2P^\top P$, the corresponding Hessians are:

$$H_1 = 2P_1^\top P_1 = 2(\mathbf{X}_{\text{lq}}\mathbf{X}_{\text{lq}}^\top) \otimes \mathbf{I}, \quad (19)$$

$$H_2 = 2P_2^\top P_2 = 2(\delta_{\text{cq}}\delta_{\text{cq}}^\top) \otimes \mathbf{I}. \quad (20)$$

Using the identity $\text{tr}(A \otimes \mathbf{I}_m) = m \cdot \text{tr}(A)$, the trace of each Hessian is:

$$\text{tr}(H_1) = 2m\|\mathbf{X}_{\text{lq}}\|_F^2, \quad \text{tr}(H_2) = 2m\|\delta_{\text{cq}}\|_F^2. \quad (21)$$

Assuming δ_{cq} and δ_{lq} are i.i.d., zero-mean quantization noises [12] and apply $\mathbf{X}_{\text{lq}} = \mathbf{X} + \delta_{\text{lq}}$:

$$\mathbb{E}[\text{tr}(H_1)] = 2m(\|\mathbf{X}\|_F^2 + \mathbb{E}\|\delta_{\text{lq}}\|_F^2), \quad (22)$$

$$\mathbb{E}[\text{tr}(H_2)] = 2m\mathbb{E}\|\delta_{\text{cq}}\|_F^2. \quad (23)$$

Since channel-wise quantization typically introduces less quantization errors, i.e., $\mathbb{E}[\|\delta_{\text{lq}}\|_F^2] > \mathbb{E}[\|\delta_{\text{cq}}\|_F^2]$ and $\|\mathbf{X}\|_F^2 > 0$, we conclude:

$$\mathbb{E}[\text{tr}(H_1)] > \mathbb{E}[\text{tr}(H_2)]. \quad (24)$$

To support this analysis, we empirically compute the trace of the Hessian in the qkv weights of Block0 in W4A4 DeiT-S. The results show that $\text{tr}(H_1) = 962.42$, while $\text{tr}(H_2) = 442.57$, which aligns with the theoretical prediction. Together with the empirical observation in Fig. 4, these results justify the stability of SOS.

It is important to note that BRECQ [19] implements a two-stage optimization strategy. In its initial stage, BRECQ conducts optimization using quantized weights alongside full-precision activations, whereas the second stage involves optimization with both being quantized. Nevertheless, despite sharing some similarities, our SOS diverges from BRECQ in three fundamental respects: 1) Motivated by the loss landscapes of ViTs, SOS first performs optimization with full-precision weights and quantized activations, while BRECQ is the opposite. As shown in Table VI of Section V-D, the order adopted by SOS plays an important role in the performance. If using the order of BRECQ, i.e., performing stage three at first, the performance suffers from degradation. 2) SOS incorporates a lossless transition specifically designed to handle high-varient activations special for ViTs, while BRECQ does not consider it. 3) BRECQ adopts the optimization method of AdaRound [51], where the weights are leaned to either be round-to-up or round-to-down. In contrast, in our method, we do not limit the learned weight's range. As shown in Table VII of Section V-D, the usage of our optimization method leads to a better performance.

In summary, stage one exploits the smooth and low-magnitude loss landscape resulting from using the channel-wise quantizer for post-LayerNorm activations and full-precision weights, ensuring a steadfast optimization process. Then, stage two realizes a lossless transition that translates the high-cost channel-wise quantizer to the efficient layer-wise quantizer for activations of all layers. Finally, stage three optimizes the model with both quantized weights and activations, further improving its performance.

V. EXPERIMENTATION

A. Experimental Settings

1) Models and Datasets: In order to demonstrate the superiority and generality of I&S-ViT, we perform extensive evaluation across diverse visual tasks, including image classification, object detection, and instance segmentation. For the image classification task, we evaluate the I&S-ViT on the ImageNet dataset [18], considering different model variants including ViT [1], DeiT [17], and Swin [32]. For object detection and instance segmentation tasks, we evaluate I&S-ViT on the COCO dataset [68] using two prevalent frameworks: Mask R-CNN [69] and Cascade Mask R-CNN [70], both with Swin [32] as the backbone.

2) *Implementation Details*: All experiments are executed utilizing the PyTorch framework [71], with pre-trained full-precision models downloaded from the Timm library. We adopt the uniform quantizer for all weights and activations except for the post-Softmax activations, which are handled by the proposed shift-uniform-log2 quantizer. We adopt the straight-through estimator (STE) [72] to bypass the calculation of the gradient of the non-differentiable rounding function. Consistent with preceding studies [14], [15], we arbitrarily select 1000 images each from the ImageNet and COCO datasets. The Adam optimizer [73] is employed for optimization. The initialized learning rate is $4e-5$ for weights, with weight decay set to 0. The learning rate undergoes adjustment via the cosine learning rate decay strategy. As pointed out in [15], [22], the quantization parameters yield numerous local minima in the loss landscape, easily misleading the learning direction. Thus, we do not optimize them after calibration. For the ImageNet dataset, the batch size is 64 and the training iteration is 200 for the W6A6 case and 1000 for other cases. For the COCO dataset, we only optimize the backbone, and the remaining structures are quantized with the calibration strategy as in [2]. A batch size of 1 with a training iteration of 1000 is used. In our experiments, SULQ's η and other quantization parameters in weight's quantizer and activation's quantizer are determined before the optimization process by grid searching for the one with the minimum mean square quantization error from predefined candidates. All experiments are implemented using a single NVIDIA 3090 GPU.

3) *Compared Methods*: We compare the proposed I&S-ViT with PTQ methods specifically designed for ViTs including PSAQ-ViT [13], Ranking-ViT [24], EasyQuant [67], PTQ4ViT [16], APQ-ViT [22], NoisyQuant [14], and RepQ-ViT [2], and PTQ methods for CNNs including BRECQ [19], QDrop [20], and PD-Quant [21], and PTQ methods for both ViTs and CNNs including Bit-shrinking [23], Bit-shrinking [23], EasyQuant [67]. In addition to the above uniform-precision methods, we further compare with the mixed-precision approach AMP-ViT [74]. We also include comparisons with several Dfq methods, such as SARDFQ [75], SMI [50], PSAQ-ViT [13], and PSAQ-ViT V2 [48]. Moreover, we report results for GPTQ [53], a method originally developed for large language models. The initial implementation of GPTQ did not perform activation quantization. For a fair comparison, we employ the activation quantizer of the state-of-the-art RepQ-ViT [2] to perform the activation quantization for GPTQ, including the reparameterization technique and the $\log\sqrt{2}$ quantizer. For other PTQ methods, we use the result reported in their paper if it exists. Otherwise, we re-implement based on their official code with the same calibration dataset as ours for a fair comparison.

B. Results on ImageNet Dataset

The comparison between the proposed I&S-ViT and other PTQ of ViTs methods is reported in Table I. Specifically, the advantages of our I&S-ViT are highlighted in all bit cases, especially for low-bit cases. As illustrated in Table I, both optimization-free and optimization-based methods suffer from non-trivial performance degradation in the ultra-low bit cases.

For instance, in the W3A3 case, optimization-based PTQ4ViT [16] suffers from collapse for all ViT variants, and RepQ-ViT presents limited accuracy. For instance, RepQ-ViT only presents 0.97%, 4.37%, and 4.84% for DeiT-T, DeiT-B, and DeiT-B, respectively. The optimization-based methods present better results but showcase an unstable performance for different ViT variants. For example, BRECQ [19] suffers from collapse on ViT-S and Swin-B. In contrast, the proposed I&S-ViT showcases a stable and considerably improved performance for ViT variants. In particular, I&S-ViT respectively presents an encouraging 40.72% and 50.68% improvement over previous methods in ViT-S and ViT-B quantization. On DeiT-T, DeiT-B, and DeiT-B, I&S-ViT respectively obtain 41.52%, 55.78%, and 73.30% performance. Compared with PD-Quant, I&S-ViT respectively presents to 1.55%, 26.45%, and 72.36% increases. On Swin-S and Swin-B, I&S-ViT reports 4.53% and 4.98% increases, respectively.

In the W4A4 case, the optimization-free RepQ-ViT and GPTQ outperform optimization-based methods on most ViT variants, demonstrating that previous optimization-based PTQ methods suffer from the overfitting issue. While the proposed I&S-ViT presents considerable improvements over RepQ-ViT across ViT variants. Specifically, compared with GPTQ, I&S-ViT achieves notable 7.17% and 5.07% improvements for ViT-S and ViT-B, respectively. When quantizing DeiT-T, DeiT-S, and DeiT-B, I&S-ViT provides notable 6.52%, 4.93%, and 6.21% accuracy gains over PD-Quant, respectively. As for Swin-S and Swin-B, I&S-ViT showcases 1.30% and 1.48% performance gains over PD-Quant, respectively.

For high-bit cases such as W5A5 and W6A6, the proposed I&S-ViT also demonstrates improved performance. In the W5A5 case, similar to the W4A4 case, optimization-based methods usually suffer a lower performance than optimization-free methods such as RepQ-ViT and GPTQ, indicating the overfitting issue. Still, I&S-ViT presents performance improvements and satisfactory results. In particular, for ViT-S and ViT-B, I&S-ViT respectively present 79.34% and 83.03% accuracy, leading to 0.73% and 0.94% increases compared with GPTQ. For DeiT-T, DeiT-S, and DeiT-B, I&S-ViT provides 0.68%, 1.22%, and 1.04% accuracy gains, respectively. As for Swin-S and Swin-B, I&S-ViT presents 82.07 and 84.05%, respectively.

In the W6A6 case, RepQ-ViT outperforms optimization-based methods for most ViT variants, indicating that optimization-based methods also suffer from the same overfitting issue as in the previous bit-width settings. Also, I&S-ViT presents performance improvements and satisfactory results. For instance, in DeiT-B, Swin-S, and Swin-B quantization, I&S-ViT presents 81.68%, 82.89%, and 84.94% accuracy, respectively, with only 0.12%, 0.34%, and 0.33% accuracy loss compared with the full-precision model.

C. Results on COCO Dataset

The results of object detection and instance segmentation are reported in Table II. It can be seen that I&S-ViT achieves a better performance in most cases. To be specific, for the W3A4 case, I&S-ViT outperforms GPTQ by 0.3 and 0.2 points on the box

TABLE II

QUANTIZATION RESULTS ON COCO DATASET. HERE, “AP^{BOX}” DENOTES THE BOX AVERAGE PRECISION FOR OBJECT DETECTION, AND “AP^{MASK}” DENOTES THE MASK AVERAGE PRECISION FOR INSTANCE SEGMENTATION. “†” INDICATES THE RESULTS ARE REPRODUCED FROM THE OFFICIAL CODES

Method	Opti.	Bit. (W/A)	Mask R-CNN				Cascade Mask R-CNN			
			w. Swin-T		w. Swin-S		w. Swin-T		w. Swin-S	
			AP ^{box}	AP ^{mask}	AP ^{box}	AP ^{mask}	AP ^{box}	AP ^{mask}	AP ^{box}	AP ^{mask}
Full-Precision	-	32/32	46.0	41.6	48.5	43.3	50.4	43.7	51.9	45.0
BRECQ [†] [19]	✓	3/4	18.9	23.2	23.6	27.8	37.4	34.3	42.7	38.1
QDrop [†] [20]	✓	3/4	8.7	10.5	30.2	31.8	21.4	19.5	21.6	19.7
PD-Quant [†] [21]	✓	3/4	16.5	18.3	27.4	29.8	35.3	31.2	41.0	36.3
GPTQ [†] [53]	✗	3/4	25.0	26.8	30.4	32.0	41.1	36.8	44.2	39.3
RepQ-ViT [†] [2]	✗	3/4	22.2	24.1	27.9	29.9	40.2	35.7	43.7	38.8
I&S-ViT (Ours)	✓	3/4	25.3	27.0	30.7	32.4	45.6	39.8	47.8	42.0
PTQ4ViT [16]	✗	4/4	6.9	7.0	26.7	26.6	14.7	13.5	0.5	0.5
APQ-ViT [22]	✗	4/4	23.7	22.6	44.7	40.1	27.2	24.4	47.7	41.1
BRECQ [†] [19]	✓	4/4	25.4	27.6	34.9	35.4	41.2	37.0	44.5	39.2
QDrop [†] [20]	✓	4/4	12.4	12.9	42.7	40.2	23.9	21.2	24.1	21.4
PD-Quant [†] [21]	✓	4/4	17.7	18.1	32.2	30.9	35.5	31.0	41.6	36.3
GPTQ [†] [53]	✗	4/4	36.3	36.3	42.9	40.2	47.1	41.5	49.2	43.2
RepQ-ViT [2]	✗	4/4	36.1	36.0	44.2 _{42.7†}	40.2 _{40.1†}	47.0	41.4	49.3	43.1
I&S-ViT (Ours)	✓	4/4	37.5	36.6	43.4	40.3	48.2	42.0	50.3	43.6
BRECQ [†] [19]	✓	6/4	33.3	32.9	39.1	37.2	43.6	38.5	45.3	39.7
QDrop [†] [20]	✓	6/4	13.6	13.0	45.3	41.3	22.1	19.6	22.2	19.6
PD-Quant [†] [21]	✓	6/4	5.1	5.0	9.0	9.3	19.1	16.7	27.0	23.9
GPTQ [†] [53]	✗	6/4	41.8	38.9	45.7	41.4	48.7	42.4	50.4	43.7
RepQ-ViT [†] [2]	✗	6/4	42.0	39.0	45.8	41.4	48.5	42.4	50.2	43.8
I&S-ViT (Ours)	✓	6/4	42.3	39.2	45.9	41.5	48.9	42.5	50.6	43.9

AP and mask AP when Mask R-CNN employs Swin-T as its backbone. If using Swin-S as its backbone, I&S-ViT improves the box AP and mask AP by 0.3 and 0.4 points, respectively. A notable gain is exhibited when it comes to Cascade Mask R-CNN. If using the Swin-T as its backbone, I&S-ViT improves the box AP and mask AP by 4.5 and 3.0 points than GPTQ, respectively. If using the Swin-S as the backbone, the proposed I&S-ViT demonstrates a 3.6 and 2.7 points on the box AP and mask AP, respectively.

For the W4A4 case, when Mask R-CNN employs Swin-T as its backbone, I&S-ViT augments the box AP and mask AP by 1.2 and 0.3 points compared with GPTQ, respectively. Similarly, with Cascade Mask R-CNN, I&S-ViT enhances the box AP by 1.1 and mask AP by 0.5 when Swin-T serves as the backbone. When Swin-S is utilized as the backbone, the improvements are 1.1 for box AP and 0.4 for mask AP.

For the W6A4 case, the proposed I&S-ViT also presents the best performance for all variants. In particular, when Mask R-CNN employs Swin-T as its backbone, I&S-ViT augments the box AP and mask AP by 0.3 and 0.2 points than RepQ-ViT, respectively. In addition, I&S-ViT augments the box AP and mask AP by 0.1 and 0.1 points if Mask R-CNN employs Swin-S as the backbone. Also, I&S-ViT respectively presents 0.4 and 0.1 points gains on the box AP and mask AP when Cascade Mask R-CNN adopts Swin-T as its backbone. If using the Swin-S as the backbone, I&S-ViT enhances the box AP by 0.4 and mask AP by 0.1, respectively.

TABLE III
ABLATION STUDIES OF THE EFFECTIVENESS OF SHIFT-UNIFORM-LOG2 QUANTIZER (SULQ) AND THE SMOOTH OPTIMIZATION STRATEGY (SOS)

Model	SULQ	SOS	Top-1 Acc. (%)
DeiT-S (W3A3)	Full-Precision		79.85
	✓		3.36
			20.70
		✓	44.79
		✓	55.78
Swin-S (W3A3)	Full-Precision		83.23
	✓		43.32
			66.50
		✓	67.77
		✓	74.20

D. Ablation Studies

To validate the effectiveness of the components within the proposed I&S-ViT, we conduct extensive ablation studies with W3A3 DeiT-S and Swin-S on the classification task.

1) *Effect of SULQ and SOS*: Table III reports the ablation study of the proposed shift-uniform-log2 quantizer (SULQ) and the smooth optimization strategy (SOS). If SULQ is not used, we utilize the uniform quantizer as an alternative. If SOS is not used, we solely conduct calibration to determine the quantization parameters of quantizers, without optimization.

TABLE IV

ABLATION STUDIES OF DIFFERENT QUANTIZERS FOR POST-SOFTMAX ACTIVATIONS. “LQ $\sqrt{2}$ ”, “LQ”, “UQ”, AND “SULQ” DENOTE THE LOG $\sqrt{2}$ QUANTIZER [2], THE LOG2 QUANTIZER, THE UNIFORM QUANTIZER, AND THE PROPOSED SHIFT-UNIFORM-LOG2 QUANTIZER, RESPECTIVELY

Model	Method	Top-1 Acc. (%)
DeiT-S (W3A3)	Full-Precision	79.85
	LQ $\sqrt{2}$	54.49
	LQ	52.60
	UQ	44.79
	SULQ (Ours)	55.78
Swin-S (W3A3)	Full-Precision	83.23
	LQ $\sqrt{2}$	67.43
	LQ	72.39
	UQ	67.77
	SULQ (Ours)	74.20

As can be observed, the proposed SULQ and SOS both contribute to the performance considerably. If both SULQ and SOS are removed, W3A3 DeiT-S and Swin-S respectively yield 3.36% and 43.32%, corresponding to 52.42% and 30.88% accuracy drop. Applying SULQ significantly improved the accuracy by 17.34% and 23.18% for DeiT-S and Swin-S, respectively. Then, by using SOS, W3A3 DeiT-S yields 44.79% accuracy and Swin-S yields 67.77% accuracy. At last, when both SULQ and SOS are adopted, it presents the best performance for both DeiT-S and Swin-S, with 55.78% for W3A3 DeiT-S and 74.20% for W3A3 Swin-S.

2) *Effect of SULQ for Post-Softmax Activations:* Table IV reports the accuracy of using different quantizers for post-Softmax activations. Here, we only replace the quantizer in our method to solely consider the superiority of SULQ.

As can be seen, if using the log $\sqrt{2}$ quantizer, W3A3 DeiT-S and W3A3 Swin-S respectively suffer from 1.29% and 9.47% accuracy degradation. When using the log2 quantizer, W3A3 DeiT-S suffers from 3.18% accuracy drops, and Swin-S suffers from 1.81% accuracy drops. If using the uniform quantizer, W3A3 DeiT-S and Swin-S suffer from 10.99% and 6.43% accuracy degradation, respectively. In contrast, the proposed SULQ presents an improved performance. In particular, SULQ achieves the highest accuracy of 55.78% for DeiT-S and 74.20% for Swin-S, indicating its superiority in handling long-tail post-Softmax activations over log $\sqrt{2}$ quantizer, log2 quantizer, and the uniform quantizer.

3) *Effect of Stages in SOS:* Table V reports the accuracy of employing different stages in SOS. By default, the SULQ is utilized, since our analysis exclusively concentrates on the stages of SOS. For the cases of not using stage one, we solely conduct calibration to determine the quantization parameters for the activation quantizer. Correspondingly, for the cases of not using stage three, we solely conduct calibration to determine the quantization parameters of the weight quantizer. If stage two is not used, we employ the layer-wise activation quantizer at first to avoid the necessity for reparameterization.

As can be seen, in the case of W3A3 DeiT-S, with no optimization stages applied, the model’s performance only presents

TABLE V

ABLATION STUDIES OF THE EFFECTIVENESS OF THE STAGES IN THE SMOOTH OPTIMIZATION STRATEGY (SOS). “S1”, “S2”, AND “S3” DENOTE STAGE ONE, STAGE TWO, AND STAGE THREE, RESPECTIVELY

Model	S1	S2	S3	Top-1 Acc. (%)
DeiT-S (W3A3)	Full-Precision			79.85
				10.99
	✓			12.17
		✓		13.41
			✓	36.95
	✓	✓		21.22
	✓		✓	45.51
		✓	✓	50.79
	✓	✓	✓	55.78
	Full-Precision			83.23
Swin-S (W3A3)				46.46
	✓			49.31
		✓		48.69
			✓	67.16
	✓	✓		49.70
	✓		✓	69.49
		✓	✓	72.90
	✓	✓	✓	74.20

10.99%. Applying stage one, stage two, and stage three increases accuracy to 12.17%, 13.41%, and 36.95%, respectively. Any combination of these three stages further yields performance gains. For example, if both stage one and stage two are employed, the accuracy increases to 21.22%, marking a substantial improvement over using either stage alone. The combination of all three stages boosts the accuracy to 55.78% with an improvement of 44.79% over no stages applied. The case of W3A3 Swin-S also exhibits the same conclusion. Individually applying stage one, stage two, and stage three increases accuracy to 49.46%, 48.69%, and 67.16%, respectively. Also, combining any two stages at a time further yields performance improvements. For example, if both stage two and stage three are employed, the accuracy increases to 72.90%. Integrating all stages leads to the highest accuracy of 74.20%, marking 27.74% over the no stages applied.

The above results reveal the contributions of each stage within the SOS. Stage one and stage two offer certain improvements. While stage three stands out as the most effective, significantly enhancing model performance when applied alone and in combination with other stages. The application of all three stages yields the highest accuracy improvements, demonstrating the effectiveness of SOS on performance.

4) *Effect of Order of SOS:* Table VI reports the accuracy of employing the order of SOS. It can be observed that the execution order of the stages in SOS is essential for the performance. If the order of stage one and stage three is reversed, W3A3 DeiT-S and Swin-S only present 16.54% and 45.87% accuracy, respectively. In contrast, when sequentially executing stage one, stage two, and stage three, W3A3 DeiT-S and Swin-S only present 55.78% and 74.20% accuracy, respectively. We attribute the observed performance gap to two primary factors. First, reversing the order of stage one and stage three introduces information loss during the transition in stage two. Specifically, after

TABLE VI

ABLATION STUDIES OF THE EFFECTIVENESS OF THE STAGES IN THE SMOOTH OPTIMIZATION STRATEGY (SOS). “S1”, “S2”, AND “S3” DENOTE STAGE ONE, STAGE TWO, AND STAGE THREE, RESPECTIVELY. “S3→S2→S1” INDICATES PERFORMING STAGE THREE, STAGE TWO, AND STAGE ONE SEQUENTIALLY. “S1→S2→S3” INDICATES PERFORMING STAGE ONE, STAGE TWO, AND STAGE THREE SEQUENTIALLY

Model	Order	Top-1 Acc. (%)
DeiT-S (W3A3)	Full-Precision	79.85
	S3→S2→S1	16.54
	S1→S2→S3	55.78
Swin-S (W3A3)	Full-Precision	83.23
	S3→S2→S1	45.87
	S1→S2→S3	74.20

TABLE VII

ABLATION STUDIES OF USING WEIGHT OPTIMIZATION METHOD OF BRECC [19] IN STAGE THREE OF SOS

Model	Optimization Method	Top-1 Acc. (%)
DeiT-S (W3A3)	Full-Precision	79.85
	BRECC [19]	36.07
	Ours	55.78
Swin-S (W3A3)	Full-Precision	83.23
	BRECC [19]	70.34
	Ours	74.20

stage three, the weights have already been quantized. Applying the transformation in (16) during stage two, where the activation quantization parameters are absorbed into the weights, alters the weight distribution. As a result, the previously obtained weight quantization parameters become invalid, leading to performance degradation. Second, initiating the process with weight quantization (i.e., applying stage three first) results in a more rugged and irregular loss landscape [65], which hinders optimization.

5) *Other Weight Optimization Method*: Table VII reports the accuracy of employing the weight optimization method of BRECC [19] to replace stage three of our SOS. Note that BRECC adopts the optimization method in AdaRound [51], which limits the optimized weight to either be round-to-up or round-to-down. In our method, we do not limit the weight’s range. It can be observed that limiting the weight’s range yields accuracy degradation. In particular, W3A3 DeiT-S and Swin-S, respectively, present 36.07% and 70.34% accuracy, corresponding to 19.71% and 3.86% drops compared with our method.

6) *Effect of Image Number*: Fig. 5 reports the ablation study of different image numbers. As can be observed, in the W3A3 DeiT-S case, when using 32 images, the top-1 accuracy is 39.70%. As the number increases, the performance improves. For 512 images, the performance is 54.13%. When it comes to 1024 images, the top-1 accuracy is 55.78%. Afterward, continually using more images does not bring a significant performance boost, as it presents only 55.89% for 2048 images. In the W3A3 Swin-S case, the top-1 accuracy is 66.04% if using 32 images. The performance also increases as the number increases. For 512 and 1024 images, the performance is 72.57% and 74.20%, respectively. It is worth noting that the proposed

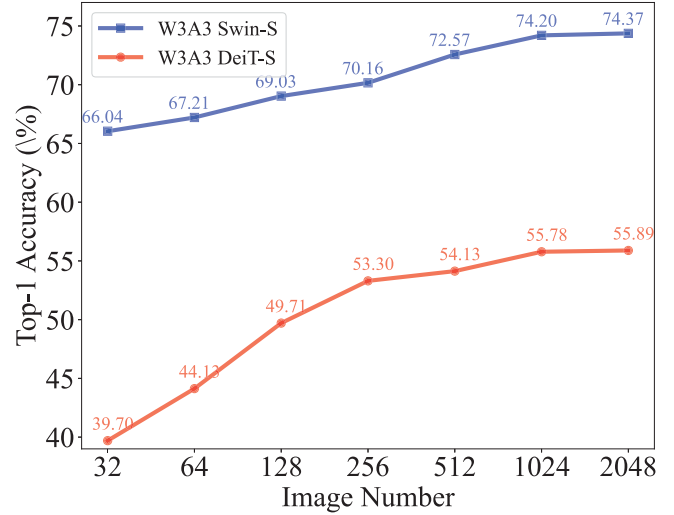


Fig. 5. The accuracy vs. image number on W3A3 DeiT-S and Swin-S.

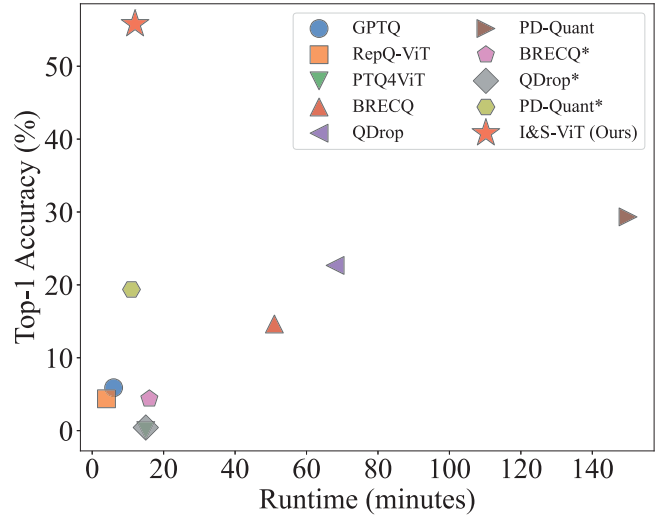


Fig. 6. The accuracy vs. runtime of PTQ methods on W3A3 DeiT-S. “*” indicates these methods are evaluated with the same iterations as ours.

I&S-ViT achieves competitive performance even with only 32 calibration images. For example, W3A3 DeiT-S presents a top-1 accuracy of 39.70%, outperforming all existing PTQ methods. As shown in Table I, the best result among these methods is 29.33%, reported by PD-Quant using 1000 calibration images. This significant performance gap, despite the use of substantially fewer samples, suggests that I&S-ViT exhibits a low risk of overfitting.

E. Time Efficiency

Fig. 6 showcases the training runtime comparison. For all methods, we use an NVIDIA 3090 GPU and set the batch size to 64. For optimization-based methods, the total number of iterations is set to match the configuration reported in their original papers. Notably, the proposed I&S-ViT significantly

TABLE VIII
QUANTIZATION RESULTS COMPARISON WITH MIX-PRECISION METHOD AND DFQ METHODS ON IMAGENET DATASET. “MP” INDICATES THE MIX-PRECISION SETTING

Method	Opti.	Bit. (W/A)	ViT-S	ViT-B	DeiT-T	DeiT-S	DeiT-B	Swin-S	Swin-B
Full-Precision	-	32/32	81.39	84.54	72.21	79.85	81.80	83.23	85.27
AMP-ViT [74]	✗	MP4/MP4	55.88	61.84	55.62	68.43	76.14	77.20	76.51
SARDFQ [75]	✓	4/4	50.32	51.84	52.06	62.29	72.17	74.74	76.42
SMI [50]	✓	4/4	24.33	35.27	30.14	42.77	65.33	65.85	65.23
PSAQ-ViT [13]	✓	4/4	47.24	36.32	47.75	58.28	71.75	73.19	71.84
PSAQ-ViT V2 [48]	✓	4/4	41.53	26.32	30.20	45.53	66.43	65.55	67.42
I&S-ViT (Ours)	✓	4/4	74.87	80.07	65.21	75.81	79.97	81.17	82.60
SARDFQ [75]	✓	5/5	74.31	70.70	66.41	74.06	78.72	79.56	80.82
SMI [50]	✓	5/5	61.33	67.53	56.44	62.88	76.74	75.41	75.25
PSAQ-ViT [13]	✓	5/5	71.59	68.48	64.10	71.90	78.45	78.15	78.50
PSAQ-ViT V2 [48]	✓	5/5	68.41	67.95	55.16	63.14	76.77	74.37	77.20
I&S-ViT (Ours)	✓	5/5	79.34	83.03	69.78	78.33	81.22	82.07	84.05
AMP-ViT [74]	✗	MP6/MP6	79.98	82.70	71.03	78.70	81.25	82.62	84.50
SARDFQ [75]	✓	6/6	78.40	79.16	69.73	77.31	80.15	80.56	83.03
SMI [50]	✓	6/6	72.95	76.33	64.03	71.65	78.81	78.25	80.30
PSAQ-ViT [13]	✓	6/6	77.20	76.65	68.37	75.85	80.00	79.74	82.00
PSAQ-ViT V2 [48]	✓	6/6	74.76	71.87	62.77	68.85	79.22	78.50	81.41
I&S-ViT (Ours)	✓	6/6	80.43	83.82	70.85	79.15	81.68	82.89	84.94

TABLE IX
QUANTIZATION RESULTS MANDARIN ASR TASK. THE CHARACTER ERROR RATE (CER) IS REPORTED AS THE METRIC

Method	Opti.	Bit. (W/A)	AiShell-1 Dev	AiShell-1 Test
Full-Precision	-	32/32	12.3	14.2
BRECQ [†] [19]	✓	6/6	30.6	32.0
QDrop [†] [20]	✓	6/6	27.3	28.8
RepQ-ViT [†] [2]	✗	6/6	15.9	18.8
GPTQ [†] [53]	✗	6/6	15.3	18.4
PD-Quant [†] [21]	✓	6/6	20.5	23.3
I&S-ViT (Ours)	✓	6/6	14.9	17.9
BRECQ [†] [19]	✓	8/8	20.6	23.9
QDrop [†] [20]	✓	8/8	18.4	21.2
RepQ-ViT [†] [2]	✗	8/8	14.9	17.2
GPTQ [†] [53]	✗	8/8	14.7	16.8
PD-Quant [†] [21]	✓	8/8	14.1	18.3
I&S-ViT (Ours)	✓	8/8	13.7	16.4

outperforms all other PTQ methods while maintaining a decent time cost. In particular, I&S-ViT roughly consumes 16 minutes. Compared with optimization-free RepQ-ViT, GPTQ, and PTQ4ViT, the consumed time of I&S-ViT remains in the same magnitude, while I&S-ViT exhibits a substantial performance improvement. When compared with optimization-based BRECQ, QDrop, and PD-Quant, the time cost of I&S-ViT is only about one-fifth to one-tenth. The time efficiency of I&S-ViT over optimization-based methods mainly comes from the fact that I&S-ViT adopts fewer total training iterations than BRECQ, QDrop, and PD-Quant. For a comprehensive comparison, we further evaluate the time costs of BRECQ, QDrop, and PD-Quant with the same iteration as ours, whose results are denoted with a star mark. As can be observed, these methods exhibit

comparable time costs to ours. However, these methods typically suffer from some performance degradation and the proposed I&S-ViT still presents significant performance gains.

To evaluate energy consumption and latency, we report GOPS/W (Giga Operations per Second per Watt) and runtime on FPGA and GPU platforms, respectively. It is important to note that both energy consumption and latency are highly dependent on the underlying hardware implementation, and substantial variation may exist across different accelerator designs [76], [77]. Therefore, we provide representative examples to demonstrate the energy efficiency and latency under different quantization settings.

GOPS/W is a standard metric for measuring computational throughput under power constraints. In our setup, a custom accelerator is implemented in Verilog, synthesized using the Vivado Design Suite, and deployed on a Xilinx Artix UltraScale+ FPGA operating at 200 MHz. The system consists of an FPGA-based compute engine, DDR4 DRAM for data buffering, and a host PC that transfers activations via Ethernet. We evaluate the energy efficiency of full-precision, W8A8, and W4A4 DeiT-S, respectively. The measured GOPS/W values are 4.12, 17.51, and 33.47, respectively, indicating that quantization improves energy efficiency. For latency evaluation, we implement quantized inference kernels using the CUTLASS library [78]. Experiments are conducted on an NVIDIA 3090 GPU using the ViT-B model, where 200 images are forwarded to measure runtime. The measured runtimes for full-precision, W8A8, and W4A4 models are 526.8 ms, 371.3 ms, and 254.1 ms, respectively, confirming that quantization yields faster inference.

F. Boarder Comparisons

1) *Comparisons With Mix-Precision and DFQ Methods:* In Table VIII, we present additional comparisons between the proposed I&S-ViT with mix-precision methods and DFQ methods.

TABLE X
QUANTIZATION RESULTS ON IMAGE SUPER-RESOLUTION TASK. THE PSNR AND SSIM [80] ARE REPORTED AS THE METRIC

Method	Model	Opti.	Bit. (W/A)	Set5 [81]	Set14 [82]	BSD100 [83]	Urban100 [84]
Full-Precision		-	32/32	38.35/0.9620	34.14/0.9227	32.44/0.9030	33.40/0.9393
BRECQ [†] [19]	SwinIR×2	✓	4/4	31.52/0.8340	29.39/0.7930	27.83/0.8094	28.75/0.7721
QDrop [†] [20]		✓	4/4	30.27/0.8914	28.98/0.8598	27.70/0.8755	28.92/0.8452
PTQ4SR [†] [85]		✗	4/4	31.16/0.8265	29.41/0.8008	28.29/0.8221	28.76/0.7756
RepQ-ViT [†] [2]		✗	4/4	29.48/0.7727	27.94/0.7389	26.48/0.7600	27.42/0.7186
GPTQ [†] [53]		✗	4/4	31.95/0.8857	29.98/0.8511	28.42/0.8662	29.70/0.8303
PD-Quant [†] [21]		✓	4/4	29.14/0.8481	28.33/0.8227	26.84/0.8359	28.50/0.8065
I&S-ViT (Ours)		✓	4/4	33.54/0.8960	30.77/0.8563	29.48/0.8794	30.09/0.8349
Full-Precision		-	32/32	32.72/0.9021	28.94/0.7914	27.83/0.7459	27.07/0.8164
BRECQ [†] [19]	SwinIR×4	✓	4/4	27.71/0.7283	25.60/0.6355	23.51/0.6341	25.40/0.6117
QDrop [†] [20]		✓	4/4	27.79/0.8187	26.02/0.7228	23.99/0.7219	25.77/0.6864
PTQ4SR [†] [85]		✗	4/4	26.10/0.6146	24.50/0.5415	22.69/0.5362	24.50/0.5231
RepQ-ViT [†] [2]		✗	4/4	28.12/0.7231	25.95/0.6366	23.95/0.6389	25.58/0.6073
GPTQ [†] [53]		✗	4/4	29.16/0.8074	26.64/0.7069	24.44/0.7106	26.21/0.6692
PD-Quant [†] [21]		✓	4/4	26.50/0.7574	25.49/0.6767	23.54/0.6733	25.44/0.6402
I&S-ViT (Ours)		✓	4/4	29.97/0.8296	27.14/0.7257	25.01/0.7370	26.53/0.6834

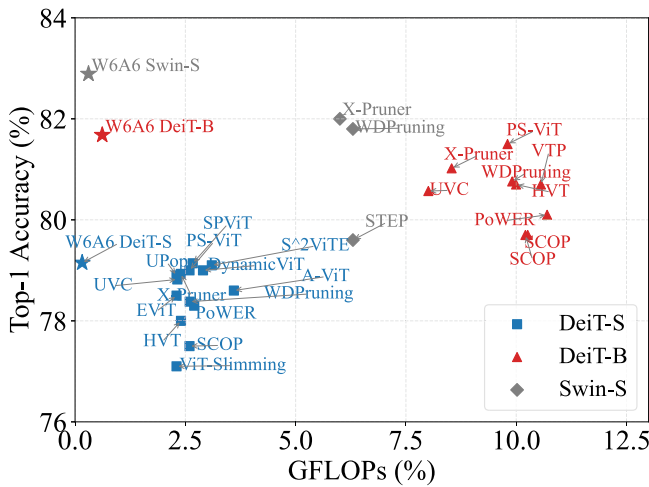


Fig. 7. The accuracy vs. GFLOPs of I&S-ViT and model pruning methods.

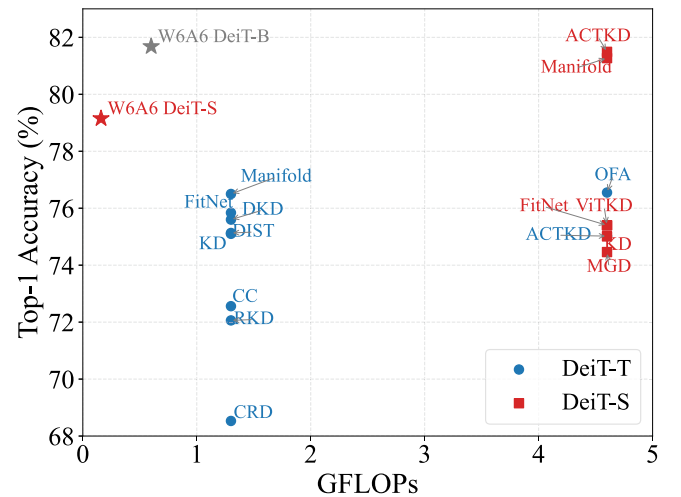


Fig. 8. The accuracy vs. GFLOPs of I&S-ViT and model distillation methods.

Notably, I&S-ViT significantly outperforms all other methods. For instance, in the W4A4 setting, the mixed-precision AMP-ViT achieves only 55.88% and 61.84% top-1 accuracy on ViT-S and ViT-B, respectively. In contrast, I&S-ViT attains 74.87% and 80.07% on the same models, demonstrating substantial performance gains. I&S-ViT also achieves notable improvements over DFQ methods across different bit-widths and ViT architectures. For example, SARDFQ reports 52.06%, 62.29%, and 72.17% accuracy on DeiT-T, DeiT-S, and DeiT-B, respectively. In comparison, I&S-ViT achieves 65.21%, 75.81%, and 79.97%, yielding improvements of 13.15%, 13.52%, and 7.80%, respectively.

2) *Comparisons With Model Pruning and Distillation Methods:* As shown in Figs. 7 and 8, we compare the proposed I&S-ViT with representative model pruning and knowledge distillation methods. For a unified and fair comparison, we adopt

GFLOPs as the efficiency metric and top-1 accuracy as the performance metric. To compute the GFLOPs of I&S-ViT, we first estimate the bit operations (BOPs) [79], and subsequently convert them into GFLOPs. It can be observed that I&S-ViT consistently achieves higher accuracy with significantly lower computational cost. This efficiency gain is primarily due to the reduced bit-widths of both weights and activations. For example, compared to model pruning methods, the W6A6 DeiT-B variant of I&S-ViT achieves a top-1 accuracy of 81.68% with only 0.6 GFLOPs, outperforming all pruning-based methods on DeiT-B. Similarly, when compared to knowledge distillation methods, the W6A6 DeiT-S model achieves 79.15% top-1 accuracy with merely 0.16 GFLOPs, offering both lower computational cost and superior performance compared to other distilled DeiT-T models.

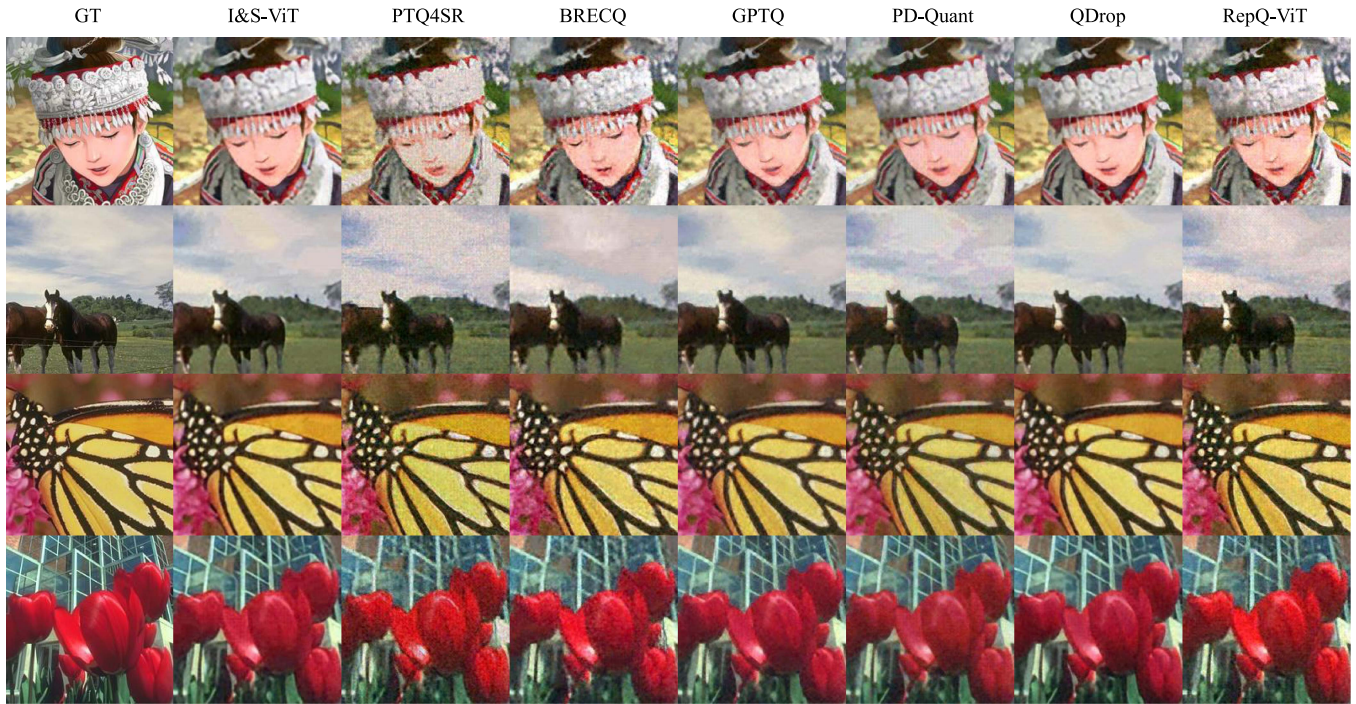


Fig. 9. Examples of qualitative comparison for image super-resolution task.

In summary, I&S-ViT not only offers better performance-efficiency trade-offs but also provides practical deployment advantages. Unlike pruning-based methods, which typically require access to the full training dataset and extensive retraining, and distillation-based methods, which rely on large pre-trained teacher models and significant computational overhead, I&S-ViT requires only a small calibration set and lightweight fine-tuning. This makes it a more practical solution for real-world deployment.

3) *Results on Speech Recognition Task*: Table IX presents the quantization results on the Mandarin automatic speech recognition (ASR) task. The training corpus consists of 8,000 hours of mobile-recorded speech data, and evaluations are conducted on the Dev and Test sets of the AiShell-1 dataset [86], containing 14,326 and 7,176 utterances, respectively. We adopt the open-source implementation of SpeechTransformer [87] to train a Transformer-based ASR model, upon which quantization is applied. It can be seen that the proposed I&S-ViT consistently outperforms existing quantization methods. For instance, under the W6A6 setting, I&S-ViT achieves character error rate (CER) improvements of 0.4 and 0.5 on the Dev and test sets of AiShell-1, respectively.

4) *Results on Image Super-Resolution Task*: Table X presents the quantization results of the image super-resolution task. It can be observed that the proposed I&S-ViT consistently outperforms other methods on both SwinIR $\times 2$ and SwinIR $\times 4$ models. For instance, on the W4A4 SwinIR $\times 2$, I&S-ViT achieves PSNR improvements of 1.59 dB, 0.79 dB, 1.06 dB, and 0.39 dB on the Set5, Set14, BSD100, and Urban100 datasets, respectively. Similarly, for the W4A4 SwinIR $\times 4$ model, I&S-ViT improves PSNR by 0.81 dB, 0.50 dB, 0.57 dB, and 0.32 dB on

the same datasets. In Fig. 9, we provide a qualitative comparison. The visual results clearly show that I&S-ViT produces the best visualization results, further demonstrating its effectiveness.

VI. CONCLUSION

In this paper, we introduced I&S-ViT, a novel optimization-based PTQ method tailored specifically for ViTs. At the outset, we address the quantization inefficiency issue associated with the log2 quantizer by introducing the shift-uniform-log2 quantizer (SULQ). The SULQ inclusively represents the full input domain to effectively address the quantization inefficiency issue and accurately approximate the distributions of post-Softmax activations. Then, our insights into the contrasting loss landscapes of different quantization granularities guide the development of the three-stage smooth optimization strategy (SOS). SOS enables stable learning by exploiting the smooth and low-magnitude loss landscape of channel-wise quantization for optimization while presenting efficiency by utilizing layer-wise quantization through seamless scale reparameterization. The superiority of I&S-ViT is demonstrated by extensive experiments on various ViTs of different vision tasks.

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